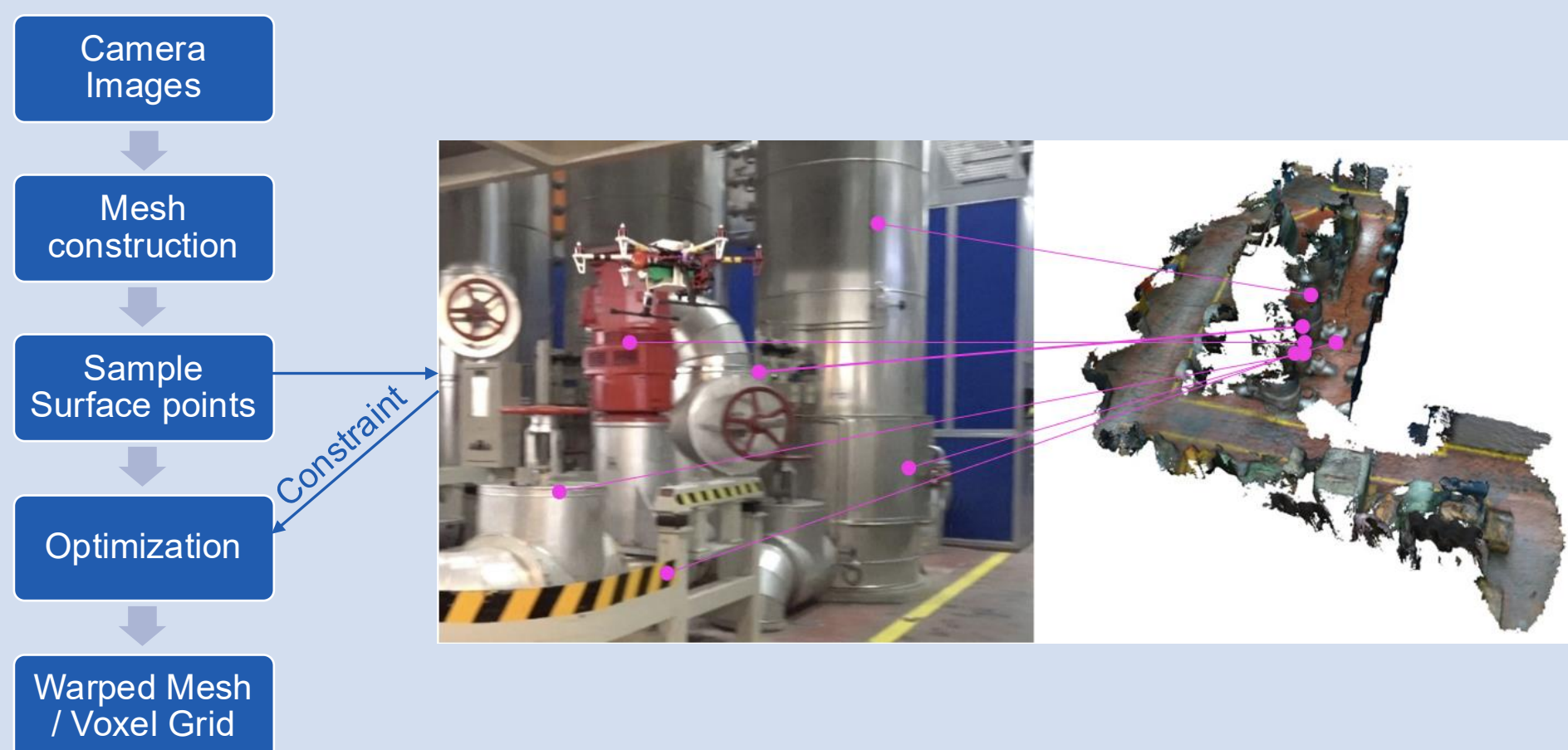


Warping Voxel Grids For 3D Robotic Mapping

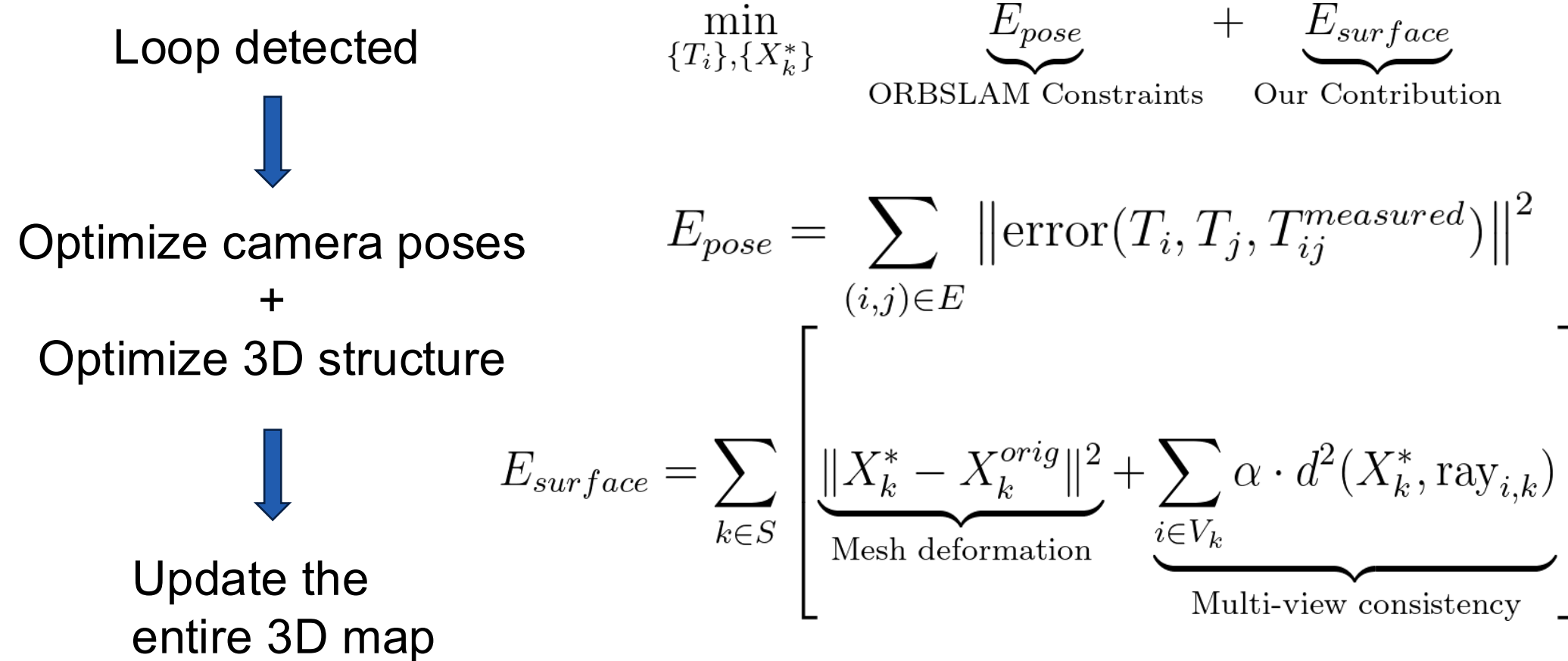
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1 Introduction

- Problem:** Traditional voxel-based 3D mapping frameworks require rebuilding the map from scratch upon loop closure
- Goal:** “Warp” the voxel grid in response to updates
- Key Idea:** Use transformations from camera images to surface points as new constraints for optimization



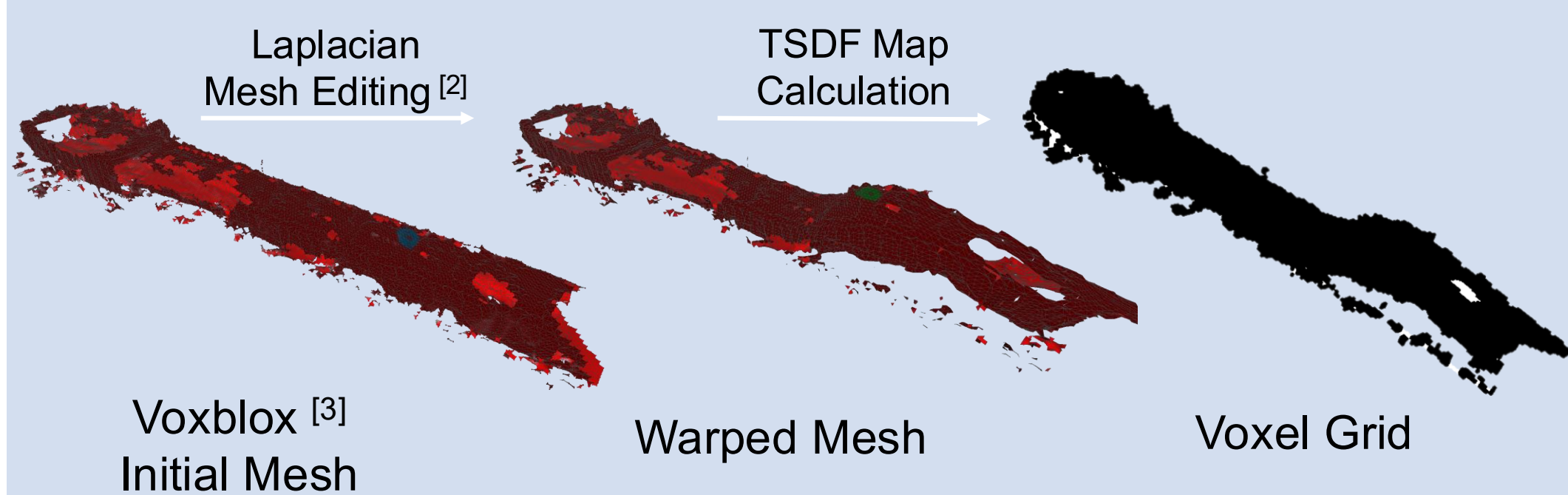
3 Combined Optimization



For loop closure optimization : Father and child frame constraints, Loop detection constraints, Shared feature constraints

For Surface-Frame Continuity Constraints (Our Contribution) : Penalizes large deformations from original mesh, Ensures 3D points project correctly to all observing cameras

4 Warping Framework

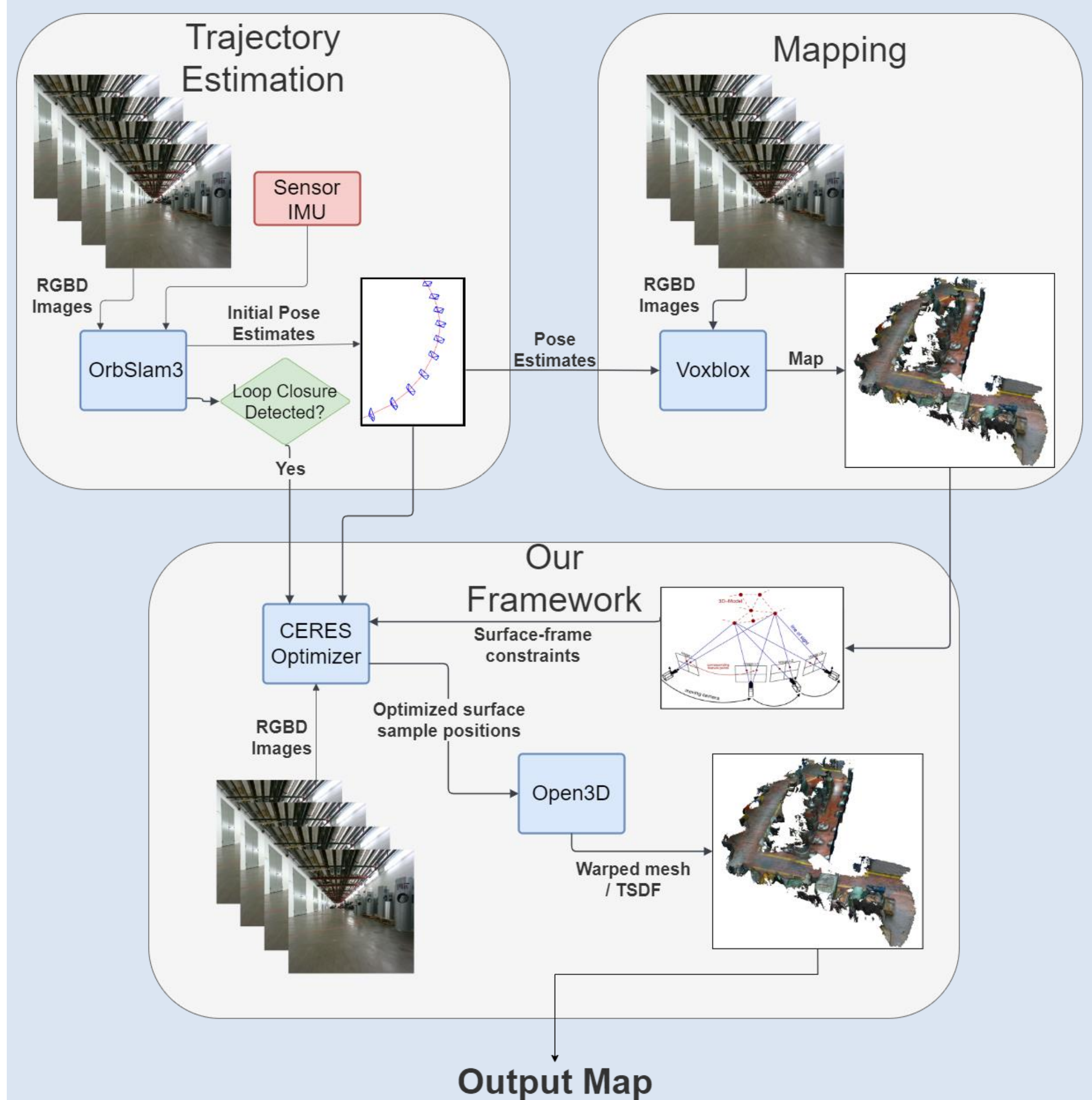


References

- [1] C. Campos, R. Elvira, J. J. Gomez, J. M. M. Montiel, and J. D. Tardós, “ORB-SLAM3: An accurate open-source library for visual, visual-inertial and multi-map SLAM,” IEEE Transactions on Robotics, vol. 37, no. 6, pp. 1874–1890, 2021
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- [3] H. Oleynikova, Z. Taylor, M. Fehr, R. Siegwart, and J. Nieto, “Voxblox: Incremental 3d euclidean signed distance fields for on-board mavplanning,” in IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2017..

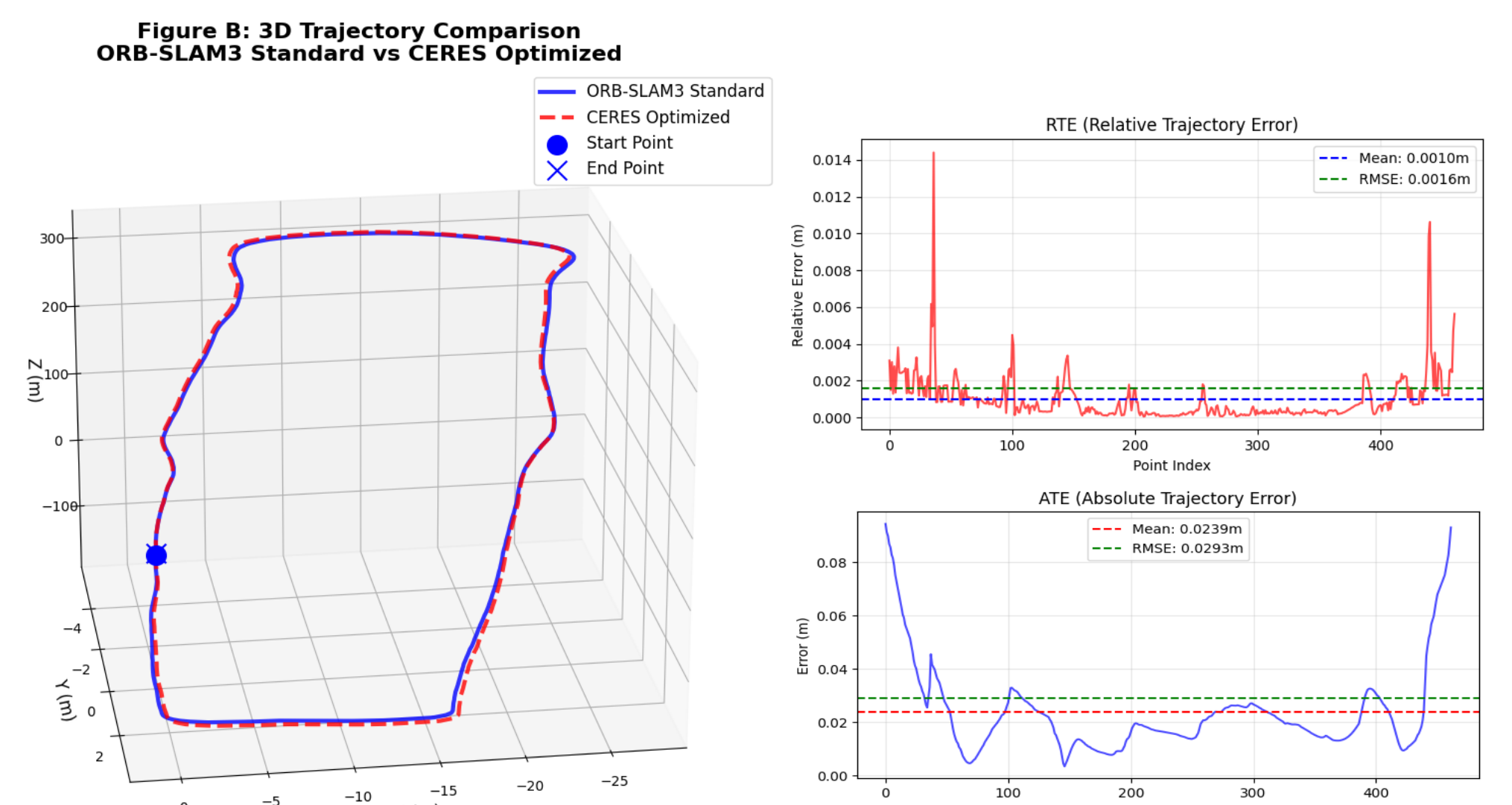
2 Method Overview

How to warp the Voxel based map?



5 Results and Future Work

- Comparison with ORB-SLAM3^[1]'s Loop Closure Completion



- Ball pivoting for surface mesh reconstruction from voxel grid

Metric	Precision	Recall	F-Score
Threshold			
0.8 * Voxel Grid Size	0.7053	0.7771	0.7395
Voxel Grid Size	0.8193	0.9073	0.8611

- Future work includes extending the optimization process to handle multiple loop closures and testing the combined pipeline.